

3 Introduction

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$$x^2 + y^2 = z^2 \quad (3.12)$$

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$$\begin{array}{c}
 y \rightarrow y^{(\ell)}, 1 \leq \ell \leq n \quad v \rightarrow v_j, 1 \leq j \leq \alpha \quad y \rightarrow y^{(\ell)}, 1 \leq \ell \leq n \\
 \downarrow \quad \quad \quad \downarrow \quad \quad \quad \downarrow \\
 -\Pi \frac{F_2 x}{\theta_1 x} \sum \frac{f_1(x, y)}{\chi' y} \frac{\delta \theta y}{\theta y} + \sum' v \frac{d^{v-1}}{dx^{v-1}} \left\{ \frac{F_2 x}{\theta_1^{(v)} x} \sum \frac{f_1(x, y)}{\chi' y} \frac{\delta \theta y}{\theta y} \right\} \quad \text{evaluate deritive at } x = \beta_j \\
 \begin{array}{c}
 F_2 x = F_0(x) \\
 \theta_1 x = f_2(x) F_0(x) \\
 f(x, y) = \frac{f_1(x, y)}{f_2(x) \chi'(y)}
 \end{array}
 \end{array}$$

$$\begin{array}{c}
 \theta_1^{(v_j)} x = v_j! \frac{f_2(x) F_0(x)}{(x - \beta_j)^{v_j}} \\
 \downarrow \\
 -\Pi \sum_{\ell=1}^n f(x, y^{(\ell)}) \frac{\delta \theta y^{(\ell)}}{\theta y^{(\ell)}} + \underbrace{\sum_{j=1}^{\alpha} \frac{1}{(v_j - 1)!} \left(\frac{d^{v_j-1}}{dx^{v_j-1}} \frac{(x - \beta_j)^{v_j}}{f_2(x)} \sum_{\ell=1}^n \frac{f_1(x, y^{(\ell)})}{\chi'(y^{(\ell)})} \frac{\delta \theta y^{(\ell)}}{\theta y^{(\ell)}} \right)}_{\text{residue at } x = \beta_j \text{ by (3.12)}} \Big|_{x=\beta_j}
 \end{array}$$

$f(x, y^{(\ell)})$

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